

1-Year Data Engineering Roadmap (Skills, Resources & Projects)

Month 1 – Advanced SQL

Skills to Learn:

- Window functions, CTEs, joins mastery, query optimization

Free Learning Resources:

- Mode SQL Tutorial
- SQLBolt Interactive Tutorial
- LeetCode SQL Practice

Project Idea:

- Practice solving 40–60 SQL problems and optimize queries on sample datasets.

Month 2 – Data Modeling

Skills to Learn:

- Star schema, snowflake schema, fact vs dimension tables, SCD

Free Learning Resources:

- Kimball Data Warehouse Guide
- Snowflake Data Warehouse Guide

Project Idea:

- Design a warehouse schema for an ecommerce or retail analytics dataset.

Month 3 – Python for Data Engineering

Skills to Learn:

- Python basics, pandas, file processing, APIs, logging

Free Learning Resources:

- freeCodeCamp Python Course
- Kaggle Pandas Course
- Python Official Tutorial

Project Idea:

- Create Python scripts that clean and load data from CSV/JSON into a database.

Month 4 – ETL Pipelines

Skills to Learn:

- ETL vs ELT, batch pipelines, ingestion pipelines, transformations

Free Learning Resources:

- Data Engineering Zoomcamp
- ETL vs ELT Article

Project Idea:

- Build a simple ETL pipeline that extracts CSV data, transforms it, and loads into a SQL warehouse.

Month 5 – Data Warehousing

Skills to Learn:

- OLTP vs OLAP, columnar storage, partitioning, analytics schemas

Free Learning Resources:

- BigQuery Documentation
- Snowflake Data Warehouse Concepts

Project Idea:

- Design a warehouse with analytics tables for product sales or marketing data.

Month 6 – Big Data Concepts

Skills to Learn:

- Distributed systems, MapReduce, big data architecture

Free Learning Resources:

- Apache Spark Documentation
- Databricks Spark Guide

Project Idea:

- Process a large dataset using PySpark and perform transformations.

Month 7 – PySpark

Skills to Learn:

- Spark DataFrames, transformations, partitioning, performance tuning

Free Learning Resources:

- Spark SQL Guide
- Databricks Learn Spark

Project Idea:

- Build a Spark ETL job that processes large logs or transaction data.

Month 8 – Workflow Orchestration

Skills to Learn:

- Airflow DAGs, scheduling pipelines, monitoring workflows

Free Learning Resources:

- Apache Airflow Documentation
- Astronomer Airflow Tutorial

Project Idea:

- Automate your ETL pipeline using Airflow scheduling.

Month 9 – Data Quality

Skills to Learn:

- Data validation, schema checks, pipeline testing

Free Learning Resources:

- Great Expectations Documentation
- Data Quality Guide

Project Idea:

- Add validation checks to ensure pipeline data quality.

Month 10 – Data Lakes

Skills to Learn:

- Lakehouse architecture, medallion architecture, Delta Lake

Free Learning Resources:

- Delta Lake Guide
- Databricks Lakehouse Guide

Project Idea:

- Build bronze, silver, gold layers for a dataset.

Month 11 – Data Engineering System Design

Skills to Learn:

- Pipeline architecture, scalability, fault tolerance

Free Learning Resources:

- Designing Data Intensive Applications Notes
- Data Mesh Architecture Article

Project Idea:

- Design architectures for analytics pipelines like Netflix-style data processing.

Month 12 – Capstone Project

Skills to Learn:

- Build an end-to-end data pipeline

Free Learning Resources:

- Data Engineering Zoomcamp Projects
- freeCodeCamp Data Engineering Project Tutorial

Project Idea:

- Build ingestion → ETL → warehouse pipeline and publish on GitHub with architecture diagram.